

Code:

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#include <stdio.h>
#include <stdlib.h>

#define MAXV 400005
#define MAXE 800005

int head[MAXV], to[MAXE], nxt[MAXE], edge_cnt;
int head_rev[MAXV], to_rev[MAXE], nxt_rev[MAXE], edge_rev_cnt;
int vis[MAXV], order[MAXV], order_idx;
int comp[MAXV], comp_cnt;
int ans[MAXV];

void link(int i, int j) {
    to[++edge_cnt] = j;
    nxt[edge_cnt] = head[i];
    head[i] = edge_cnt;

    to_rev[++edge_rev_cnt] = i;
    nxt_rev[edge_rev_cnt] = head_rev[j];
    head_rev[j] = edge_rev_cnt;
}

void dfs1(int u) {
    vis[u] = 1;
    for (int e = head[u]; e; e = nxt[e]) {
        int v = to[e];
        if (!vis[v]) dfs1(v);
    }
    order[++order_idx] = u;
}

void dfs2(int u, int c) {
    comp[u] = c;
    for (int e = head_rev[u]; e; e = nxt_rev[e]) {
        int v = to_rev[e];
        if (!comp[v]) dfs2(v, c);
    }
}

int main() {
```

```

int n, m;
if (scanf("%d %d", &n, &m) != 2) return 0;

for (int i = 0; i < n; ++i) {
    char s1, s2;
    int u, v;
    scanf(" %c%d %c%d", &s1, &u, &s2, &v);

    int pos_u = (s1 == '+') ? u : u + m;
    int neg_u = (s1 == '-') ? u + m : u;
    int pos_v = (s2 == '+') ? v : v + m;
    int neg_v = (s2 == '-') ? v + m : v;

    link(neg_u, pos_v);
    link(neg_v, pos_u);
}

int total_nodes = 2 * m;
for (int i = 1; i <= total_nodes; ++i) {
    if (!vis[i]) dfs1(i);
}

for (int i = total_nodes; i >= 1; --i) {
    int u = order[i];
    if (!comp[u]) {
        dfs2(u, ++comp_cnt);
    }
}

for (int i = 1; i <= m; ++i) {
    if (comp[i] == comp[i + m]) {
        printf("IMPOSSIBLE\n");
        return 0;
    }
    ans[i] = comp[i] > comp[i + m];
}

for (int i = 1; i <= m; ++i) {
    printf("%c%c", ans[i] ? '+' : '-', (i == m) ? '\n' : ' ');
}

return 0;
}

```

Input:

3 5
+1 +2
-1 +3
+4 -2

Output:

- + + + -